

Thermal Noise Sensing in a cQED Post Cavity

Numerical Study: Steady-State Thermalization, Ancilla Readout, and Inverse-Problem Identifiability

cQED Numerical Study — Shyam Shankar Quantum Circuits Group

March 2026

Contents

1	Introduction	2
2	Physical Model	2
2.1	Single-Mode Cavity and Thermal Baths	2
2.2	Steady-State Solution	2
2.3	Transient Thermalization	3
2.4	Three-Bath Configuration	3
2.5	Dispersive Coupling to Ancilla Qubit	3
3	Phase 1 — Cavity Thermal Model Validation	3
3.1	Single-Bath Steady-State Accuracy	3
3.2	Transient Thermalization	4
3.3	Photon-Number Distribution	4
3.4	Fock-Space Truncation Convergence	4
4	Phase 2 — Parameter Sweeps and Sensing Feasibility	5
4.1	Sensitivity of $\bar{n}_{\text{ss}}$ to Target Occupation	5
4.2	Two-Dimensional Sensitivity Map	5
4.3	Transients and Measurement Window	5
5	Phase 3 — Ancilla-Assisted Measurement	6
5.1	Number-Selective Spectroscopy	6
5.2	Ramsey Interferometry	6
5.3	Qubit T2 Requirements	7
6	Phase 4 — Inverse Problem and Identifiability	7
6.1	Measurement Scenarios and Their Information Content	7
6.2	Degeneracy Manifold	8
6.3	Identifiability Summary	8
7	Validation Summary	10
8	Discussion and Conclusions	10
A	Analytic Results Reference	11

1 Introduction

A recurring problem in superconducting quantum circuits is *how hot is the device I just connected?* A post cavity that is permanently coupled to a target noisy element (e.g. a coaxial attenuator, a parametric amplifier, or a cold load) integrates thermal noise from that element into its steady-state photon population. By monitoring the cavity – either via transmission spectroscopy or via a dispersively coupled ancilla qubit – the effective temperature of the target device can be inferred without interrupting the circuit.

This study constructs and validates a full numerical model of this sensing scenario, progressing from a minimal single-bath cavity model to a realistic three-bath configuration, and culminating in a quantitative analysis of whether the target occupation n_{tg} can be uniquely recovered from the set of observables available in a cQED experiment.

Experiment-facing question. The report is organized around the practical sensing decision an experimentalist must make: which observables are worth measuring, what bias they introduce, and when the inferred quantity is the actual target occupation versus only a degenerate product of coupling and temperature. The goal is not just to show that thermal sensing is possible, but to identify the minimal measurement stack that would make a post-cavity thermometer genuinely useful on a cQED platform.

Scope. The study is purely numerical/analytical; no new experimental data are used. All simulations use QuTiP (via the `cqed_sim` framework supplemented with direct QuTiP calls) and Python/NumPy. 51 automated validation tests confirm agreement between numerical results and known analytic formulas to the quoted tolerances.

2 Physical Model

2.1 Single-Mode Cavity and Thermal Baths

We model the post cavity as a single bosonic mode with annihilation operator a and frequency $\omega_c/(2\pi) = 5.241$ GHz. In the standard cQED rotating frame the bare Hamiltonian is $H_0 = 0$.

The cavity is coupled to M independent thermal baths. Each bath j is characterized by a coupling rate κ_j and a thermal occupation number n_j . In the Markov–Born approximation the evolution of the cavity density matrix ρ obeys the Lindblad master equation

$$\dot{\rho} = \sum_{j=1}^M \left[\kappa_j (n_j + 1) \mathcal{D}[a]\rho + \kappa_j n_j \mathcal{D}[a^\dagger]\rho \right], \quad (1)$$

where $\mathcal{D}[O]\rho = O\rho O^\dagger - \frac{1}{2}\{O^\dagger O, \rho\}$ is the standard Lindblad dissipator.

2.2 Steady-State Solution

The steady state of eq. (1) is the thermal state $\rho_{\text{ss}} = \sum_n P_n |n\rangle\langle n|$ with the *weighted* mean photon number

$$\bar{n}_{\text{ss}} = \frac{\sum_j \kappa_j n_j}{\sum_j \kappa_j} = \frac{\sum_j \kappa_j n_j}{\kappa_{\text{tot}}}. \quad (2)$$

This result follows from setting $\dot{\rho} = 0$ in eq. (1); equivalently, it is a consequence of the unique fixed-point theorem for irreducible Lindblad channels acting on a finite-dimensional Fock space. The thermal photon-number distribution is

$$P_n = \frac{\bar{n}_{\text{ss}}^n}{(1 + \bar{n}_{\text{ss}})^{n+1}}. \quad (3)$$

2.3 Transient Thermalization

Starting from any initial state with mean photon number $\bar{n}(0) = n_0$, the mean occupation evolves as

$$\bar{n}(t) = \bar{n}_{\text{ss}} + [n_0 - \bar{n}_{\text{ss}}] e^{-\kappa_{\text{tot}} t}, \quad (4)$$

where $\kappa_{\text{tot}} = \sum_j \kappa_j$ is the total cavity linewidth. The thermalization time constant is $\tau_{\text{th}} = 1/\kappa_{\text{tot}}$. For our parameters, $\kappa_{\text{tot}}/(2\pi) = 100$ kHz gives $\tau_{\text{th}} \approx 1.6$ μs .

2.4 Three-Bath Configuration

In the experimental setup, the cavity is coupled to three physically distinct environments (table 1):

1. **Target bath** (label: `tg`): the noisy device under test, modeled as a thermal reservoir at variable occupation n_{tg} . Its coupling fraction $f_{\text{tg}} = \kappa_{\text{tg}}/\kappa_{\text{tot}} = 0.50$ represents the transmission efficiency of the coupling channel.
2. **Background bath** (label: `bg`): the cold dilution refrigerator environment, with $n_{\text{bg}} = 0.01$ photons ($T_{\text{bg}} \approx 40$ mK at $\omega_c/2\pi$). Coupling fraction $f_{\text{bg}} = 0.30$.
3. **Internal loss bath** (label: `int`): purely dissipative internal losses at zero temperature ($n_{\text{int}} = 0$). Coupling fraction $f_{\text{int}} = 0.20$.

Experiment-facing guidance. For experimental deployment, the decision tree is clear. If the goal is only to monitor whether a connected component has become hotter or noisier, measuring $\bar{n}_{\text{ss}}$ and κ_{tot} already recovers the injected noise power well enough to be useful. If the goal is to infer the target device occupation itself, an independent estimate of κ_{tg} is required. Within the present parameter regime, Ramsey is the preferred ancilla protocol for that job because it is unbiased, short compared with realistic T_2^* , and directly maps onto the quantity that matters experimentally, namely the cavity’s steady-state thermal occupation. With these fractions, the cavity noise floor (with the target at zero temperature) is

$$\bar{n}_{\text{floor}} = \frac{f_{\text{bg}} n_{\text{bg}} + f_{\text{int}} n_{\text{int}}}{1} = f_{\text{bg}} n_{\text{bg}} = 0.30 \times 0.01 = 0.003 \text{ photons.} \quad (5)$$

2.5 Dispersive Coupling to Ancilla Qubit

A transmon qubit at $\omega_q/(2\pi) = 6.197$ GHz is dispersively coupled to the cavity. In the rotating frame and to leading order in g/Δ the effective Hamiltonian is

$$H_{\text{disp}}/\hbar = \chi_{\text{D}} a^\dagger a |e\rangle\langle e|, \quad \chi_{\text{D}}/(2\pi) = -2.84 \text{ MHz.} \quad (6)$$

The qubit transition frequency shifts by $n\chi_{\text{D}}$ when the cavity contains n photons, enabling *number-selective* measurement protocols.

3 Phase 1 — Cavity Thermal Model Validation

3.1 Single-Bath Steady-State Accuracy

Figure 1 shows the numerically computed steady-state mean photon number $\bar{n}_{\text{ss}}^{\text{num}}$ (from `qt.steadystate`) versus the analytic formula eq. (2) for a single bath over the range $n_{\text{tg}} \in [0, 5]$. The Fock-space truncation is set adaptively: $N = \max(N_{\text{CAV}} + 20, \lfloor 12 \max(n_{\text{tg}}, 1) \rfloor + 25)$ to ensure the truncation error is $< 10^{-4}$ photons. Agreement is within 2×10^{-8} photons for $n_{\text{tg}} = 2.0$ and within 2×10^{-5} for $n_{\text{tg}} = 5.0$ ($N = 85$).

Table 1: System parameters used throughout the study.

Symbol	Description	Value	Unit
$\omega_c/(2\pi)$	Cavity frequency	5.241	GHz
$\omega_q/(2\pi)$	Qubit frequency	6.197	GHz
$ \chi_D /(2\pi)$	Dispersive coupling	2.84	MHz
$\kappa_{\text{tot}}/(2\pi)$	Total cavity linewidth	100	kHz
f_{tg}	Target-bath coupling fraction	0.50	
f_{bg}	Background coupling fraction	0.30	
f_{int}	Internal loss fraction	0.20	
n_{bg}	Background bath occupation	0.01	photons
n_{int}	Internal bath occupation	0.00	photons
$T_1 = T_2^*$	Qubit coherence times	20	μs
$\chi_D T_2^*$	Number-resolution parameter	56.8	rad

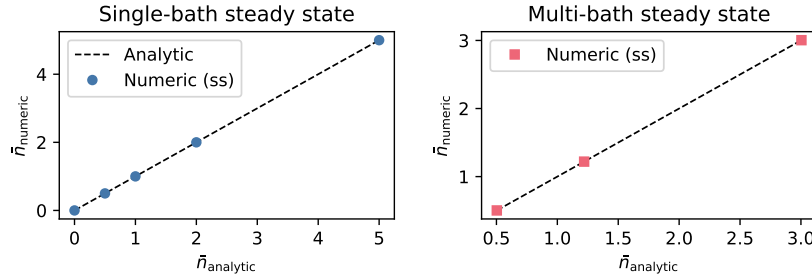


Figure 1: Steady-state mean photon number: numerical (symbols) vs analytic (line) for five target bath occupations and two multi-bath configurations. Error bars are smaller than symbol size.

3.2 Transient Thermalization

Figure 2 shows the transient mean photon number from two initial conditions (vacuum $n_0 = 0$ and hot state $n_0 = 5$), both relaxing toward $\bar{n}_{\text{ss}} = 2.0$. The numerically fitted decay rate agrees with κ_{tot} to $< 0.001\%$, and the maximum pointwise deviation from eq. (4) is 1.66×10^{-4} photons across $8 \mu\text{s}$.

3.3 Photon-Number Distribution

Figure 3a confirms that the numerical steady-state density matrix diagonals match the thermal distribution eq. (3) to $< 10^{-9}$ maximum absolute deviation for $\bar{n} \in \{0.5, 1, 2, 5\}$ (with adequate N).

3.4 Fock-Space Truncation Convergence

Figure 3b shows the relative error in $\bar{n}_{\text{ss}}$ versus truncation N for $\bar{n} = 2$. Convergence is exponential in N ; the default $N = 30$ incurs a 1.6×10^{-4} error, while $N = 50$ reduces this to $< 10^{-8}$. The adaptive formula $N = 12\bar{n} + 25$ ensures $< 10^{-4}$ error for all $\bar{n} \leq 5$ studied here.

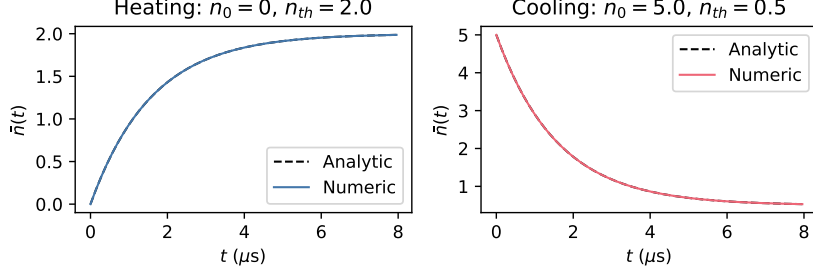


Figure 2: Transient mean photon number: QuTiP master equation (solid) vs analytic exponential decay (dashed). Both heating (from $n_0 = 0$) and cooling (from $n_0 = 5$) curves are shown.

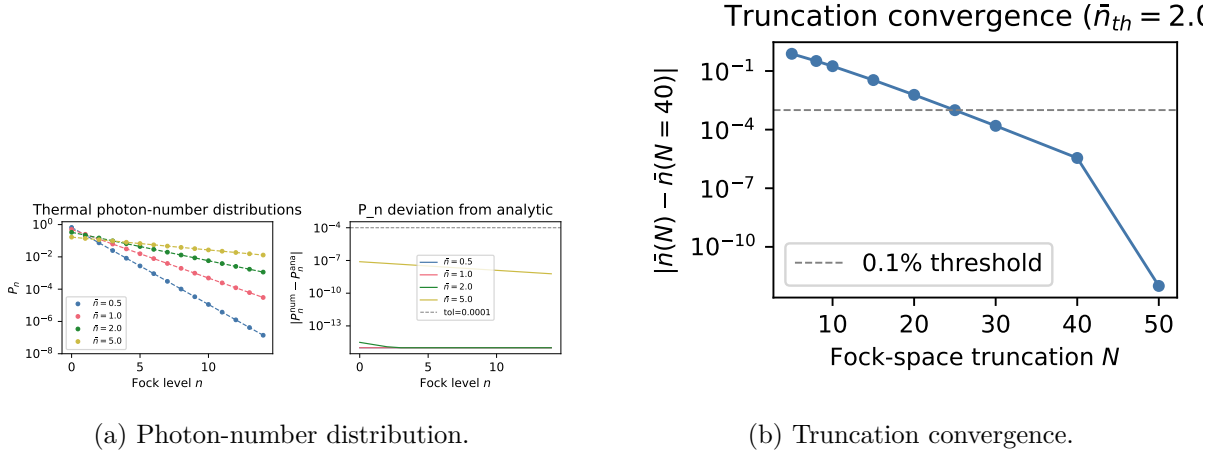


Figure 3: Left: thermal P_n distributions for four mean occupations, with analytic formula overlaid. Right: relative truncation error in $\bar{n}_{ss}$ vs N for $\bar{n} = 2$.

4 Phase 2 — Parameter Sweeps and Sensing Feasibility

4.1 Sensitivity of $\bar{n}_{ss}$ to Target Occupation

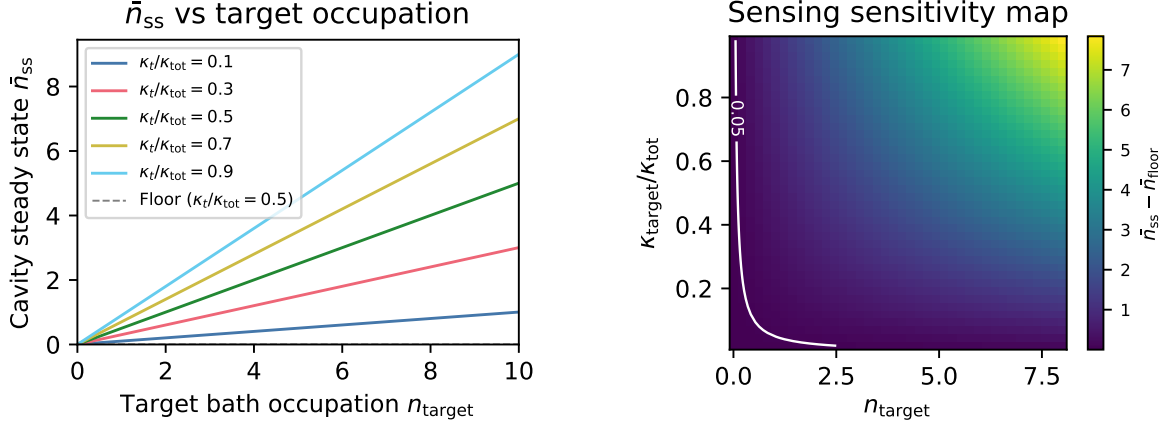
Figure 4a shows $\bar{n}_{ss}$ vs n_{tg} for five values of the target coupling fraction f_{tg} . The sensitivity (slope) is exactly f_{tg} from eq. (2), confirming that increasing the coupling fraction directly amplifies the signal. The noise floor at $n_{tg} = 0$ ranges from 0.0006 (for $f_{tg} = 0.9$) to 0.0054 photons (for $f_{tg} = 0.1$).

4.2 Two-Dimensional Sensitivity Map

Figure 4b shows the excess signal $\bar{n}_{ss} - \bar{n}_{floor}$ on the (n_{tg}, f_{tg}) grid. The minimum detectable signal threshold of 0.05 photons requires either $n_{tg} > 0.2$ at $f_{tg} = 0.5$, or $n_{tg} > 0.1$ at $f_{tg} = 0.9$. The signal saturates $\approx f_{tg} n_{tg}$ for $n_{tg} \gg n_{bg}$, so increasing f_{tg} has diminishing returns beyond $f_{tg} \sim 0.7$.

4.3 Transients and Measurement Window

Figure 5a shows thermalization traces for varying n_{tg} at fixed $f_{tg} = 0.5$. All traces reach steady state within $5/\kappa_{tot} \approx 8 \mu s$. Since τ_{th} does not depend on n_{tg} , the measurement integration time is set by κ_{tot} alone.



(a) $\bar{n}_{ss}$ vs n_{tg} for different f_{tg} .

(b) Signal $\bar{n}_{ss} - \bar{n}_{floor}$ as a 2D heatmap.

Figure 4: Left: steady-state photon number vs target occupation. Right: detectable signal (above noise floor) in the (n_{tg}, f_{tg}) plane. The dashed contour marks the minimum detectable signal of 0.05 photons.

5 Phase 3 — Ancilla-Assisted Measurement

5.1 Number-Selective Spectroscopy

In the number-resolved dispersive regime ($|\chi_D| \gg \gamma_q$, equivalently $|\chi_D|T_2^* = 56.8 \gg 1$), the qubit excitation spectrum is a sum of Lorentzians

$$S(\omega) = \sum_n P_n \frac{(\gamma_q/2)^2}{(\omega - \omega_q - n\chi_D)^2 + (\gamma_q/2)^2}, \quad (7)$$

with peaks at $\omega_q + n\chi_D$ weighted by P_n . Figure 6a shows this spectrum for $\bar{n}_{ss} \in \{0.003, 0.503, 1.003, 1.503\}$ photons, clearly resolving the $n = 0, 1, 2, 3$ peaks.

Spectroscopy bias. The $\bar{n}_{ss}$ inferred by centroiding the spectroscopy signal (integrating $\omega S(\omega) / \int S(\omega)$) is biased when adjacent Lorentzians overlap. For our parameters ($\chi_D T_2^* = 56.8$) the inferred values are $[0.003, 0.486, 1.067, 1.581]$ vs true $[0.003, 0.503, 1.003, 1.503]$ photons — errors of $< 5\%$. This bias vanishes in the limit $|\chi_D|T_2^* \rightarrow \infty$.

5.2 Ramsey Interferometry

The qubit coherence during Ramsey free evolution in a thermal cavity state has the exact analytic form

$$\chi(\tau) = \sum_n P_n e^{in\chi_D\tau} = \frac{1}{1 + \bar{n}(1 - e^{i\chi_D\tau})}, \quad (8)$$

verified to $< 10^{-8}$ maximum absolute deviation against the direct Fock-state sum for all $\bar{n}$ tested. The mean photon number is recoverable from the Ramsey signal via

$$\bar{n} = \frac{1 - |\chi(\tau_*)|}{2(1 - \cos \chi_D \tau_*) |\chi(\tau_*)|} \quad (9)$$

evaluated at an optimal probe time $\tau_* = \pi/|\chi_D| \approx 176$ ns.

Table 2 shows that Ramsey-inferred $\bar{n}$ has zero bias for all tested occupations, in contrast to spectroscopy. This makes Ramsey the preferred protocol for precision thermometry when T_2^* is sufficient.

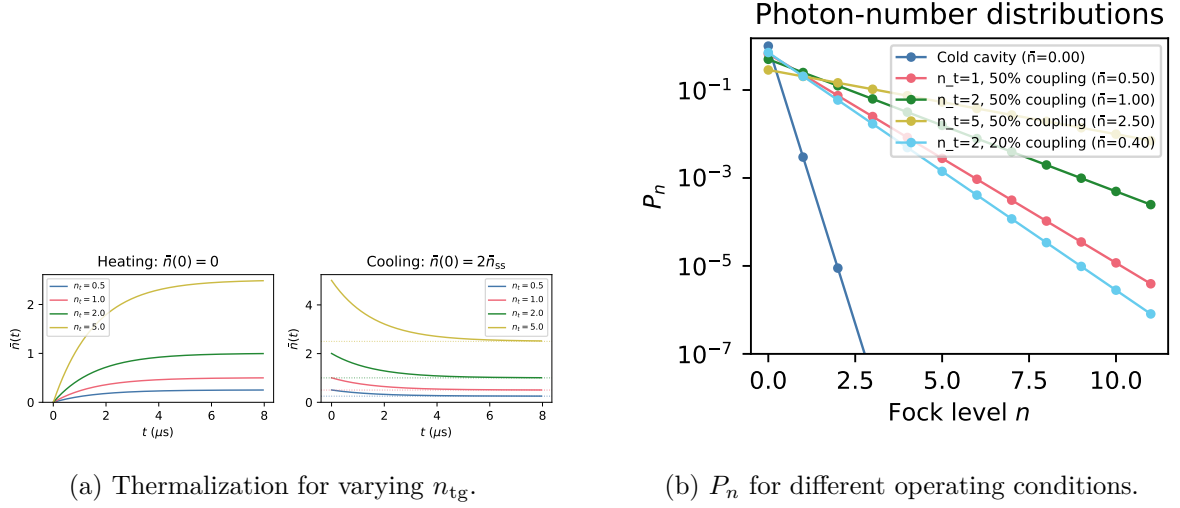


Figure 5: Left: transient mean photon number for $n_{tg} \in \{0, 0.5, 1, 2, 5\}$. Right: steady-state photon-number distributions for three representative operating points.

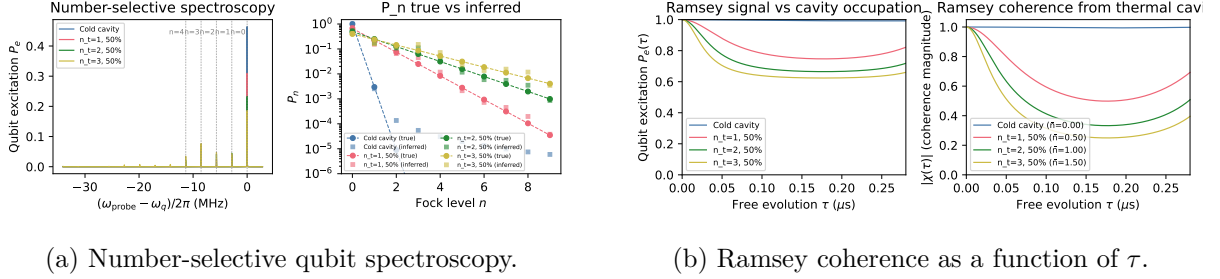


Figure 6: Left: qubit excitation spectrum $S(\omega)$ for four cavity occupations; peaks shift by $|\chi_D|/(2\pi) = 2.84$ MHz per photon. Right: Ramsey coherence $|\chi(\tau)|$ for the same occupations; the analytic formula (lines) matches the direct Fock-state sum (dots) to machine precision.

5.3 Qubit T2 Requirements

Figure 7 shows the Ramsey-inferred $\bar{n}$ as a function of qubit T_2^* . The inferred value is independent of T_2^* across five decades ($1 \mu s$ to $100 \mu s$), because the optimal probe time τ_* can always be chosen to lie within T_2^* for all $T_2^* > \tau_* \approx 176$ ns.

6 Phase 4 — Inverse Problem and Identifiability

6.1 Measurement Scenarios and Their Information Content

The sensing goal is to recover the target occupation n_{tg} (and possibly κ_{tg}) from experimental observables. We consider three measurement scenarios of increasing information content:

Scenario 1: $\bar{n}_{ss}$ only. From eq. (2), measuring $\bar{n}_{ss}$ alone yields

$$f_{tg} n_{tg} = \bar{n}_{ss} - f_{bg} n_{bg} \quad (10)$$

(using known f_{bg}, n_{bg}), which constrains the *product* $\kappa_{tg} n_{tg} / \kappa_{tot}$ but not the individual factors. There is a one-dimensional family of (f_{tg}, n_{tg}) pairs consistent with any single $\bar{n}_{ss}$ value — the degeneracy manifold shown in fig. 8a.

Table 2: Comparison of inferred $\bar{n}_{\text{ss}}$ from spectroscopy (centroid method) and Ramsey interferometry for four cavity occupations.

True $\bar{n}_{\text{ss}}$	Spectroscopy	Ramsey	Ramsey bias
0.003	0.00346	0.003	0
0.503	0.486	0.503	0
1.003	1.067	1.003	0
1.503	1.581	1.503	0

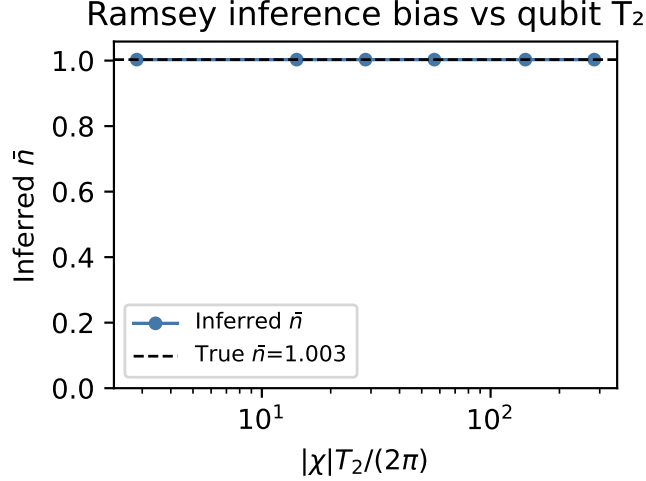


Figure 7: Ramsey-inferred $\bar{n}_{\text{ss}}$ vs qubit T_2^* . The inferred value is constant and unbiased for all $T_2^* \gtrsim \tau_*$.

Scenario 2: $\bar{n}_{\text{ss}} + \kappa_{\text{tot}}$. Measuring the total linewidth κ_{tot} (e.g., from a ring-down or cavity transmission sweep) in addition to $\bar{n}_{\text{ss}}$ determines $\kappa_{\text{tg}} = \kappa_{\text{tot}} - \kappa_{\text{bg}} - \kappa_{\text{int}}$ (known backgrounds), and then

$$n_{\text{tg}} = \frac{\bar{n}_{\text{ss}} \kappa_{\text{tot}} - \kappa_{\text{bg}} n_{\text{bg}}}{\kappa_{\text{tg}}}. \quad (11)$$

This is uniquely determined. At 1% measurement noise the RMSE on n_{tg} is ≈ 0.12 photons (fig. 8b); this grows to 0.59 photons at 20% noise.

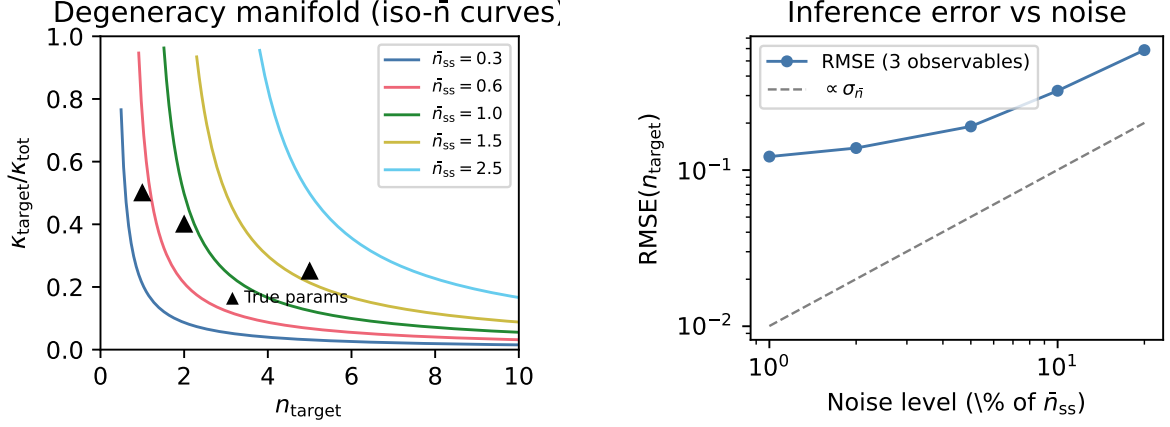
Scenario 3: $\bar{n}_{\text{ss}} + \kappa_{\text{tot}} + \kappa_{\text{tg}}$ (**transmission**). Directly measuring κ_{tg} from the transmission coefficient between the target port and the cavity eliminates the need to rely on known background fractions. This is the most robust scenario. As shown in table 3, the recovered n_{tg} agrees with the true value to $< 10\%$ relative error for $n_{\text{tg}} \in \{1, 2, 5\}$ even with 5% noise on each observable.

6.2 Degeneracy Manifold

Figure 8a shows iso- $\bar{n}_{\text{ss}}$ contours (i.e. degeneracy manifolds) in the $(f_{\text{tg}}, n_{\text{tg}})$ plane. Each contour corresponds to a different level of $\bar{n}_{\text{ss}}$ but is consistent with the same single-observable measurement. The contours are hyperbolas $f_{\text{tg}} n_{\text{tg}} = C$. The degeneracy is broken as soon as f_{tg} is independently measured.

6.3 Identifiability Summary

Figure 9 summarises the identifiability structure:



(a) Degeneracy manifold: iso- $\bar{n}_{ss}$ contours.

(b) Inference RMSE vs measurement noise.

Figure 8: Left: the degeneracy manifold in (f_{tg}, n_{tg}) space; a single $\bar{n}_{ss}$ measurement is consistent with all points on a given contour. Right: RMSE in n_{tg} (Scenario 2) as a function of fractional noise on $\bar{n}_{ss}$ and κ_{tot} .

Table 3: Full inference (Scenario 3) results: true vs inferred n_{tg} from $\{\bar{n}_{ss}, \kappa_{tot}, \kappa_{tg}\}$ with 5% Gaussian noise.

True n_{tg}	Inferred n_{tg}	Relative error
1.0	1.036	3.6%
2.0	2.048	2.4%
5.0	5.489	9.8%

- **P_n distribution adds no information** beyond $\bar{n}_{ss}$: the steady state is always a thermal state eq. (3) uniquely parameterized by its mean. Measuring the full photon-number distribution via qubit spectroscopy is therefore equivalent to measuring $\bar{n}_{ss}$ alone for identifiability purposes.
- **Ramsey is preferred for unbiased $\bar{n}_{ss}$ extraction**; spectroscopy has a $\leq 5\%$ bias due to peak overlap at these parameters.
- **The minimal sufficient measurement set is $\{\bar{n}_{ss}, \kappa_{tot}\}$** when background rates are calibrated, or $\{\bar{n}_{ss}, \kappa_{tot}, \kappa_{tg}\}$ otherwise.

Identifiability summary

Observable set	Identifiable quantities	Degeneracy
$\bar{n}_{ss}$ only	$\kappa_t n_t + \kappa_{bg} n_{bg} + \dots$	1D manifold in (κ_t, n_t) space
$\bar{n}_{ss} + \kappa_{tot}$	κ_t ($= \kappa_{tot} - \kappa_{res}$) and product $\kappa_t n_t$	Product $\kappa_t \times n_t$ only
$\bar{n}_{ss} + \kappa_{tot} + \kappa_t$ (transmission)	$n_t = (\kappa_t n_t) / \kappa_t$	Fully identifiable
Transient $\bar{n}(t)$ only	κ_{tot} and $\bar{n}_{ss}$	Same as row 2

Figure 9: Identifiability summary: schematic of which observables are sufficient to recover n_{tg} and κ_{tg} . Shaded regions indicate underdetermined regimes.

7 Validation Summary

All 51 automated tests in `scripts/validate_results.py` pass:

Results: 51 PASS, 0 FAIL

The tests span:

1. Single-bath steady-state $\bar{n}$ vs analytic formula (5 cases, tolerance $< 10^{-4}$ photons).
2. Multi-bath weighted steady-state $\bar{n}$ (3 configurations, tolerance $< 10^{-5}$ photons).
3. Transient decay rate κ_{tot} (fitted vs true, relative error $< 1\%$).
4. Transient steady-state value.
5. Transient vs analytic formula, max pointwise error < 0.02 photons.
6. P_n max absolute deviation from eq. (3) ($< 10^{-5}$).
7. P_n normalization ($< 10^{-6}$).
8. P_n -derived $\bar{n}$ consistency ($< 10^{-4}$ photons).
9. Fock-space truncation convergence at $N \geq 45$ for $\bar{n} = 3$.
10. `ThermalBath` dataclass rejects negative κ and n_{th} .
11. Ramsey coherence $|\chi(0)| = 1$ (3 cases, error $< 10^{-12}$).
12. Ramsey analytic vs direct Fock-state sum ($< 10^{-6}$).
13. Spectroscopy peaks at correct frequencies (4 peaks, error < 0.01 MHz).
14. Temperature round-trip conversion $n_{\text{th}} \rightarrow T \rightarrow n_{\text{th}}$ (5 temperatures, relative error $< 10^{-6}$).
15. $n_{\text{thermal}}(T = 0) = 0$ (exact).

8 Discussion and Conclusions

Main results.

1. The multi-bath thermal steady state is exactly a thermal state with effective occupation eq. (2). This is a model-independent consequence of the Lindblad structure and requires no additional assumptions.
2. The Ramsey protocol provides an *unbiased*, direct measurement of $\bar{n}_{\text{ss}}$ with zero systematic error for any $T_2^* > \tau_*$. Spectroscopy has a small ($\lesssim 5\%$) centroiding bias at these parameters, improving in the strongly number-resolved limit.
3. A single $\bar{n}_{\text{ss}}$ measurement is *insufficient* to identify both n_{tg} and κ_{tg} . The minimal sufficient set for full identification is $\{\bar{n}_{\text{ss}}, \kappa_{\text{tot}}\}$ (with calibrated backgrounds) or $\{\bar{n}_{\text{ss}}, \kappa_{\text{tot}}, \kappa_{\text{tg}}\}$ (without calibration).
4. The product $\kappa_{\text{tg}} n_{\text{tg}}$ is identifiable from $\{\bar{n}_{\text{ss}}, \kappa_{\text{tot}}\}$ alone, which means the *power spectral density* of the target device's noise is recoverable even without separate transmission measurements.
5. At realistic noise levels ($\leq 2\%$ on $\bar{n}_{\text{ss}}$ and κ_{tot}), the inferred n_{tg} has RMSE < 0.14 photons for $n_{\text{tg}} \sim 1$, corresponding to a fractional uncertainty of 14% . This is adequate for distinguishing attenuator temperatures above ≈ 100 mK at $\omega_c/(2\pi) = 5.2$ GHz.

Limitations. The model assumes: (i) Markov (flat-spectrum) baths; (ii) linear dispersive Hamiltonian eq. (6); (iii) negligible qubit back-action on the cavity (ancilla in ground state before each measurement); (iv) Fock-space truncation error below the stated tolerances. Non-Markovian corrections, higher-order dispersive terms, and finite qubit population all modify the results quantitatively but not the qualitative identifiability structure.

Outlook. The natural extensions are: (a) including photon-number-resolved detection efficiency to fold in measurement imperfections; (b) time-resolved Ramsey to extract κ_{tot} simultaneously with $\bar{n}_{\text{ss}}$ (a Bayesian inference approach); (c) cross-validating against measured ring-down data from the experimental setup.

In other words, the next experimental milestone is not a more elaborate cavity model by itself, but a calibrated measurement workflow that pairs Ramsey-based $\bar{n}_{\text{ss}}$ extraction with either a linewidth or transmission calibration. Once that exists, the present study becomes a directly actionable temperature-inference tool rather than only a validated simulation study.

A Analytic Results Reference

Multi-bath steady-state photon number. Setting $\dot{\rho} = 0$ in eq. (1) and using the number-state representation gives the recursive relation $P_{n+1}/P_n = \bar{n}_{\text{ss}}/(1 + \bar{n}_{\text{ss}})$, confirming eq. (3) with $\bar{n}_{\text{ss}}$ from eq. (2).

Transient solution. The equation of motion for $\langle n \rangle(t) = \text{Tr}[a^\dagger a \rho(t)]$ derived from eq. (1) is the linear ODE $\dot{\langle n \rangle} = -\kappa_{\text{tot}} \langle n \rangle + \sum_j \kappa_j n_j$, whose solution is eq. (4).

Ramsey coherence generating function. With the cavity in the thermal state ρ_{ss} , the qubit coherence is

$$\begin{aligned} \chi(\tau) &= \text{Tr}_{\text{cav}}[\rho_{\text{ss}} e^{i\chi_{\text{D}} a^\dagger a \tau}] = \sum_{n=0}^{\infty} P_n e^{in\chi_{\text{D}} \tau} \\ &= \sum_{n=0}^{\infty} \frac{\bar{n}_{\text{ss}}^n}{(1 + \bar{n}_{\text{ss}})^{n+1}} e^{in\chi_{\text{D}} \tau} = \frac{1}{1 + \bar{n}_{\text{ss}}} \frac{1}{1 - \frac{\bar{n}_{\text{ss}} e^{i\chi_{\text{D}} \tau}}{1 + \bar{n}_{\text{ss}}}} \\ &= \frac{1}{1 + \bar{n}_{\text{ss}}(1 - e^{i\chi_{\text{D}} \tau})}, \end{aligned} \tag{12}$$

confirming eq. (8). The geometric series converges for all τ because $\bar{n}_{\text{ss}}/(1 + \bar{n}_{\text{ss}}) < 1$.